

Yerrabandla Sai Pranav Reddy

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LinkedIn

GitHub

Portfolio

Education

Neil Gogte Institute of Technology

Bachelor of Engineering in Computer Science and Engineering — **CGPA: 9.004**

2023 – Present

Hyderabad, Telangana

Experience

DrugParadigm

April 2025 – Present

AI Research Intern, Onsite

Hyderabad, Telangana

- Architected and deployed structure-aware generative and regression models for peptide and enzyme modalities using Graph Neural Networks (GNNs), Geometric Deep Learning, and Diffusion models in PyTorch. Engineered end-to-end training and inference pipelines on NVIDIA DGX multi-GPU systems, optimizing memory utilization for large-scale biological datasets. **Co-authored** a manuscript published at Briefings in Bioinformatics (**Oxford University Press**).
- InversePep – Peptide Inverse Folding (Generative Modeling)**: Developed a structure-conditioned generative model for peptide inverse folding, trained on a massive dataset of **38,000+ sequences** and structures. Integrated Transformer encoders, conditional Diffusion models, and GVP-based Graph Neural Networks (GVP-GNN). Achieved a **0.51 TM-score**, a **18%+** improvement over ProteinMPNN(0.31) and ESM-IF1(0.33) benchmarks.
- GraphEnzyme – Enzyme Kinetics Prediction (Regression Modeling)**: Designed a multimodal framework for enzyme kcat prediction using **23,000+ sequences**, structures, and SMILES strings. Fused GVP-GNNs for structures, GCNs for substrate molecules, and ChemBERTa for SMILES with Squeeze-and-Excitation (SE) blocks for adaptive weighting, achieving **PCC 0.81** and **R² 0.64** with **5%+** improved consistency.

Publications

InversePep: Diffusion-driven structure-based inverse folding for functional peptides

- Briefings in Bioinformatics, Oxford University Press · Q1 Journal · DOI: 10.1093/bib/bbag277*

Projects

HireGraph - Multi-Agent AI Hiring Assistant

Live Demo | GitHub

- Built a multi-agent hiring assistant with 7 specialized **LangGraph** agents across 3 LLMs (**Groq 8B** for parallel screening, **Groq 70B** for ranking, **Gemini 2.5 Flash** for critique) with automated evaluation pipeline distinguishing project-proven skills from listed skills across unstructured resume data.
- Architected a self-correcting **Judge-Executor** critique loop where **Gemini** audits **Groq's** rankings via verdict-based retry and **team-aware ranking** that measures each candidate's marginal value against existing team skill gaps.
- Built full-stack with **React + Vite**, **FastAPI**, **Clerk JWT auth**, and real-time SSE pipeline visualization with **prompt engineering**-driven agent orchestration deployed on Render and Vercel with exponential backoff rate limiting.

NyayLens - Transformer-Based Legal QA System

Live Demo | GitHub

- Architected an end-to-end Legal AI platform over **26K+ Supreme Court judgments (298K+ embeddings)** with a **FastAPI Agentic Orchestrator** autonomously routing across Document-QA, Hybrid RAG, and Summarization pipelines for intent-aware legal research.
- Engineered a **Hybrid RAG** engine combining **FAISS dense search + SQLite FTS5**, Reciprocal Rank Fusion, and BGE cross-encoder reranking with a prompt-level Approved Source Registry — structurally eliminating LLM hallucinations and enforcing citation-only responses.
- Built a decoupled full-stack system (**React/Vercel + FastAPI/HuggingFace Spaces**) leveraging **Legal-BERT + Legal-PEGASUS** for local summarization and **Llama 3.1 8B via Groq** for sub-second conversational QA, hardened with thread-safe SQLite and async lifecycle management.

Bloom - AI-Powered Plant Disease Detection

Live Demo | GitHub

- Built an plant disease classification system on **87k+ images** across **38 classes** – Implemented CNN architectures from scratch in NumPy and production-grade TensorFlow models, achieving **92% accuracy** and **97% accuracy** respectively.
- Deployed a production-ready system with frontend (Vercel), backend APIs (Render), MongoDB for user and prediction management and model hosting on HuggingFace for real-time inference.

Technical Skills

- Programming**: Python, Java, JavaScript, C
- Core CS**: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
- Web Development**: Node.js, Express.js, REST APIs, React, Angular, MySQL, MongoDB, PostgreSQL, HTML, CSS
- Machine Learning & AI**: PyTorch, TensorFlow, Neural Networks (ANNs CNNs), Transformers (BERT, T5), Graph Neural Networks (GNNs GCNs GVP), Diffusion Models
- Tools & Systems**: Git, Docker, Postman, VS Code

Achievements and Certifications

- Ranked in the **Top 5%** among 50,000+ participants in the Amazon ML Challenge (2025).
- Secured **3rd place** in a 12-hour internal hackathon at Keshav Memorial Engineering College (KMEC).